frequency, but with a new amplitude and a new phase shift. In equation form, the two representations are related:

EQUATION 8-5
The addition of a cosine and sine wave results in a cosine wave with a different amplitude and phase shift. The information contained in A & B is transferred to two other variables, M and θ.

$$A\cos(x) + B\sin(x) = M\cos(x + \theta)$$

The important point is that no information is lost in this process; given one representation you can calculate the other. In other words, the information contained in the amplitudes A and B, is also contained in the variables M and θ. Although this equation involves sine and cosine waves, it follows the same conversion equations as do simple vectors. Figure 8-9 shows the analogous vector representation of how the two variables, A and B, can be viewed in a rectangular coordinate system, while M and θ are parameters in polar coordinates.

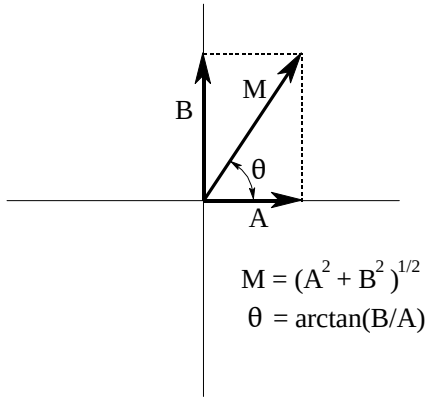

FIGURE 8-9
Rectangular-to-polar conversion. The addition of a cosine wave and a sine wave (of the same frequency) follows the same mathematics as the addition of simple vectors.

In polar notation, $Mag\,X[\ ]$ holds the amplitude of the cosine wave (M in Eq. 8-4 and Fig. 8-9), while $Phase\,X[\ ]$ holds the phase angle of the cosine wave (θ in Eq. 8-4 and Fig. 8-9). The following equations convert the frequency domain from rectangular to polar notation, and vice versa:

EQUATION 8-6
Rectangular-to-polar conversion. The rectangular representation of the frequency domain, $Re\,X[k]$ and $Im\,X[k]$, is changed into the polar form, $Mag\,X[k]$ and $Phase\,X[k]$.

$$Mag\,X[k] = (Re\,X[k]^2 + Im\,X[k]^2)^{1/2}$$

$$Phase\,X[k] = \arctan\left(\frac{Im\,X[k]}{Re\,X[k]}\right)$$

EQUATION 8-7
Polar-to-rectangular conversion. The two arrays, $Mag\,X[k]$ and $Phase\,X[k]$, are converted into $Re\,X[k]$ and $Im\,X[k]$.

$$Re\,X[k] = Mag\,X[k]\ \cos(Phase\,X[k])$$

$$Im\,X[k] = Mag\,X[k]\ \sin(Phase\,X[k])$$